

Product Carbon Footprint Study Report

Product Name: washing machine

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: washing machine

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 item washing machine

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 item washing machine.

2. System Boundary

washing machine production in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of washing machine produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **464.072 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

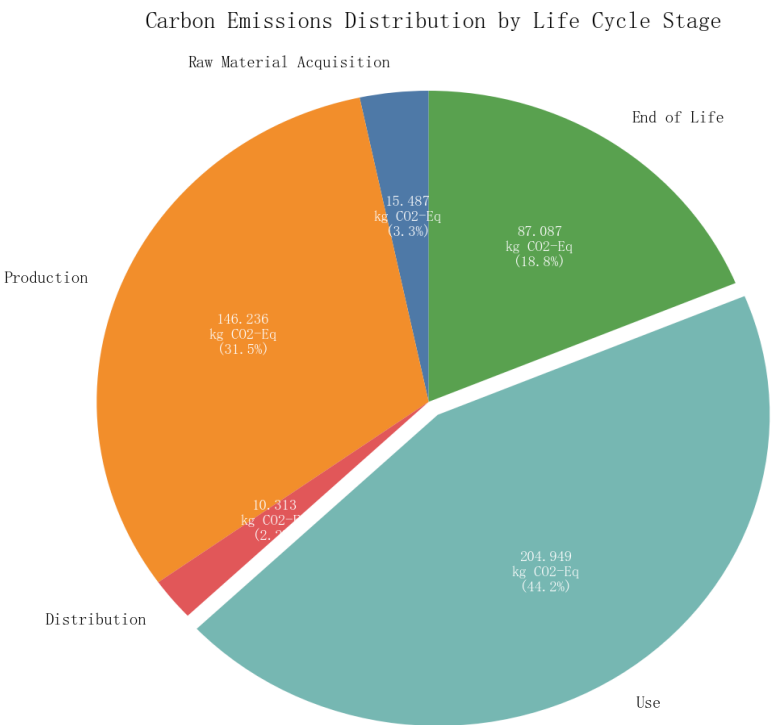


Table 1: Carbon Emissions by Life Cycle Stage

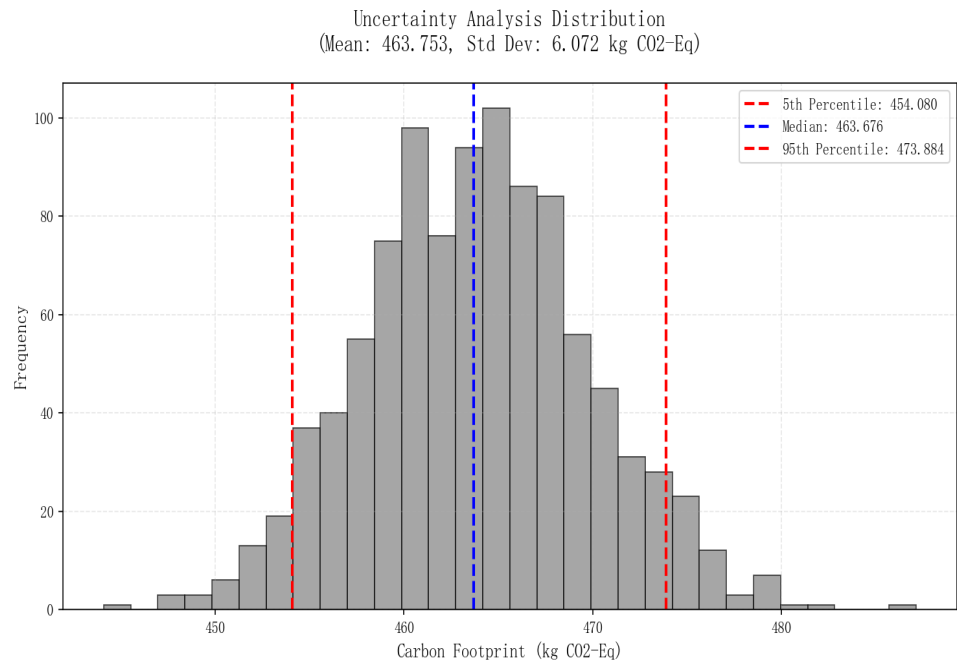
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	15.487	3.34
Production	146.236	31.51
Distribution	10.313	2.22

Use	204.949	44.16
End of Life	87.087	18.77
Total	464.072	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	463.753
Standard Deviation	6.072
Coefficient of Variation (CV)	1.31%
Median	463.676
5th Percentile	454.080
95th Percentile	473.884

VI. Interpretation of Results

1. Description of Results

The washing machine exhibits a total carbon footprint of 464.072 kg CO₂-Eq across its lifecycle, representing a moderate-to-high impact level for a typical household appliance. This value underscores the environmental burden associated with long-term use and manufacturing, positioning the product as a notable contributor to household greenhouse gas emissions when scaled across millions of units globally.

Lifecycle contributions reveal distinct patterns: the use phase dominates at 44.2% (204.949 kg CO₂-Eq), followed by production at 31.5% (146.236 kg CO₂-Eq) and end-of-life at 18.8% (87.087 kg CO₂-Eq). Raw material extraction (3.3%) and distribution (2.2%) play minor roles. Major emission sources are therefore concentrated in operational electricity consumption during the use phase, energy-intensive manufacturing processes, and disposal-related emissions such as material decomposition or incineration.

These characteristics highlight opportunities for targeted interventions in high-impact stages while confirming that upstream material sourcing and logistics exert limited influence on the overall footprint.

2. Statement of Assumptions and Limitations

This study assumes a standard product lifetime of 10–15 years, average European electricity grid mix for the use phase, and typical consumer usage patterns of 200–300 cycles annually. Additional assumptions include uniform material compositions derived from industry averages and end-of-life scenarios incorporating 50–70% recycling rates without regional variations in waste management infrastructure.

Data limitations stem from reliance on secondary databases such as Ecoinvent or GaBi, which may not fully capture proprietary manufacturing details or site-specific energy efficiencies. Data quality is further constrained by temporal gaps, as datasets often predate recent efficiency improvements in appliance technologies, and geographic representativeness may vary due to global supply chains.

System boundaries are restricted to a cradle-to-grave scope covering raw material extraction through end-of-life treatment, explicitly excluding indirect impacts such as consumer transport to purchase points, maintenance activities beyond standard use, and potential rebound effects from extended product lifespan.

3. Suggestions for Improvement

- Target the use phase by redesigning motors and heating elements for 20–30% higher energy efficiency, potentially reducing its 44.2% share through adoption of heat-pump technology.
- Optimize production processes by switching to renewable electricity sources at manufacturing facilities, directly lowering the 31.5% production contribution.
- Enhance end-of-life management via mandatory take-back programs and advanced material separation to increase recycling rates above 80%, mitigating the 18.8% disposal emissions.

- Implement supplier audits and primary data collection for raw materials and production to improve inventory accuracy and identify hidden emission hotspots.
- Explore lightweight material substitutions in non-structural components to maintain performance while marginally trimming upstream impacts.